Software Requirements Specification

Ghana Climate Resilience and Flood Intelligence Programme Website

Phase 1 – Public Information and Partnership Website

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1. Project Overview

The Ghana Climate Resilience and Flood Intelligence Programme (GCRFIP) website will serve as the official public-facing digital platform for introducing the programme, communicating its vision, presenting its strategic documents, attracting collaborators, and supporting partnership engagement.

The website is not intended, in Phase 1, to function as a live flood intelligence dashboard, data platform, or operational decision-support system. Instead, it will act as a professional credibility platform for government institutions, universities, development partners, NGOs, private-sector organizations, and the general public.

The primary objective is to present GCRFIP as a serious, well-structured, nationally relevant climate resilience programme.

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2. Website Purpose

The website shall:

- introduce GCRFIP clearly and professionally;
- explain Ghana's flood challenge;
- present the programme vision, mission, philosophy, and six pillars;
- showcase the programme architecture and implementation roadmap;
- provide access to key documents and infographics;
- invite expressions of interest from potential partners;
- support communication with government, academia, development partners, NGOs, private sector, and communities;
- establish a professional online identity for GCRFIP.

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3. Target Users

The website shall serve the following user groups:

Government Institutions

Users from ministries, agencies, MMDAs, NADMO, GMet, Hydrological Services Department, Water Resources Commission, and related institutions.

Development Partners and Donors

Users from the World Bank, African Development Bank, UNDP, GCF, USAID, GIZ, JICA, EU, foundations, and other climate/development organizations.

Universities and Research Institutions

Researchers, professors, postgraduate students, research centres, and international academic collaborators.

NGOs and Civil Society Organizations

Organizations involved in disaster risk reduction, climate adaptation, public health, sanitation, environmental protection, and community engagement.

Private Sector and Technology Partners

Companies involved in AI, GIS, IoT, cloud computing, software development, satellite data, drones, telecommunications, engineering, and infrastructure.

General Public

Citizens interested in flood resilience, environmental stewardship, climate adaptation, and community participation.

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4. Project Scope

Included in Phase 1

The Phase 1 website shall include:

- public informational pages;
- responsive design;
- downloadable documents;

- contact form;
- partnership expression-of-interest form;
- newsletter signup form;
- news/insights section;
- basic search or filtering for resources if feasible;
- SEO metadata;
- analytics integration;
- deployment to public hosting platform.

Excluded from Phase 1

The following are not required in Phase 1:

- user login system;
- database-backed admin dashboard;
- live flood data dashboard;
- GIS map dashboard;
- real-time sensor integration;
- payment system;
- user accounts;
- operational flood warning system;
- partner portal;
- mobile application;
- multilingual support, unless later requested.

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5. Recommended Technical Stack

Front-End

Recommended:

- Next.js
- React
- TypeScript
- Tailwind CSS

Alternative acceptable stack:

- Astro

- TypeScript
- Tailwind CSS

The final choice should prioritize performance, maintainability, clean design, and easy deployment.

Back-End

For Phase 1, no full backend is required.

Simple serverless functions or form service integrations may be used for:

- contact form;
- partnership expression-of-interest form;
- newsletter signup.

Form Handling

Recommended options:

- Formspree
- Netlify Forms
- Getform
- Tally embed
- Google Forms embed, only if a very simple setup is preferred

Preferred implementation:

- custom-designed frontend forms connected to a simple form service;
- submissions forwarded to the official GCRFIP email address;
- optional CSV export through the form service dashboard.

Database

No database is required for Phase 1.

The website may later be upgraded to use:

- Supabase;
- PostgreSQL;
- Firebase;
- Strapi;
- Directus;
- Sanity CMS.

Hosting

Recommended:

- Vercel for Next.js;
- Netlify for static or Astro/Next.js deployment.

The developer shall prepare the website for deployment with:

- custom domain support;
- HTTPS;
- environment variables;
- production build configuration.

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6. Website Structure

The website shall include the following core pages.

6.1 Home Page

Purpose

The Home page shall provide a strong first impression and immediately communicate what GCRFIP is, why it matters, and how users can engage.

Required Content Sections

The Home page shall include:

1. Hero section
    - Programme name
    - Tagline: “Intelligence Today. Resilience Tomorrow.”
    - Short description
    - Primary call-to-action: “Explore the Programme”
    - Secondary call-to-action: “Partner With Us”
2. The Flood Challenge
    - Brief summary of Ghana’s flood risk challenge

    - Link to full Flood Challenge page
3. The GCRFIP Solution
    - Short explanation of the integrated national programme
    - Mention six pillars
4. Six Pillars Preview
    - Six cards, one for each pillar
    - Each card links to the Six Pillars page or individual pillar section
5. National Impact
    - Short benefit statements:
        - safer communities;
        - stronger institutions;
        - better flood intelligence;
        - improved public health;
        - climate resilience.
6. Documents Preview
    - Executive Portfolio
    - Programme Prospectus
    - Founder Roadmap
    - Blueprint Summary
7. Partnership Callout
    - Invitation to government, universities, NGOs, private sector, and development partners
8. Footer
    - Quick links
    - Contact email
    - Social media links
    - Copyright notice

6.2 About GCRFIP Page

Purpose

Explain the identity, vision, mission, philosophy, and origin of GCRFIP.

Required Content Sections

- About the programme
- Vision
- Mission
- Programme motto
- Programme philosophy:
  - Prevent
  - Understand
  - Monitor
  - Predict
  - Protect
  - Strengthen
- Founder message
- Why the programme was created
- Long-term ambition

6.3 The Flood Challenge Page

Purpose

Explain why Ghana needs GCRFIP.

Required Content Sections

- Flooding as a national development challenge
- Drivers of increasing flood risk:
  - extreme rainfall;
  - rapid urbanization;
  - inadequate drainage;
  - environmental degradation;

    - development in flood-prone areas;
    - weak coordination;
    - public health risks.
- Impacts of flooding:
    - loss of lives;
    - infrastructure damage;
    - disease outbreaks;
    - economic losses;
    - displacement;
    - environmental degradation.
- "Why now?" section
- Figure or infographic area
- Link to Partnership page

6.4 The Six Pillars Page

Purpose

Present the six pillars of GCRFIP.

Required Content Sections

Each pillar shall have a section containing:

- title;
- short description;
- why it matters;
- flagship initiatives;
- expected benefits.

Pillars

1. Community Flood Monitoring and Prevention
2. Scientific Research and Flood System Understanding
3. Smart Flood Intelligence and Decision Support
4. Public Health, Sanitation and Community Resilience
5. Governance and Institutional Strengthening

6. Capacity Building and Knowledge Transfer

6.5 Programme Architecture Page

Purpose

Explain how the six pillars work together as one integrated national system.

Required Content Sections

- National Flood Intelligence Platform concept
- Flood Intelligence Digital Twin concept
- Data sources:
  - communities;
  - sensors;
  - satellites;
  - weather stations;
  - hydrological models;
  - GIS;
  - public health data;
  - infrastructure data.
- Data flow:
  - observe;
  - integrate;
  - analyse;
  - forecast;
  - decide;
  - act;
  - learn.
- Simplified programme ecosystem diagram
- Explanation of how GCRFIP supports decision-making

6.6 Roadmap Page

Purpose

Present the founder roadmap and national implementation roadmap.

Required Content Sections

- Founder Roadmap
  - Build foundation
  - Establish identity
  - Build advisory board
  - Build academic partnerships
  - Build government partnerships
  - Build technology ecosystem
  - Engage communities and civil society
  - Establish founding consortium
  - Launch programme
  - Pursue strategic funding
- National Implementation Roadmap
  - Phase 1: Programme establishment and pilot
  - Phase 2: Regional expansion
  - Phase 3: National rollout
  - Phase 4: Continuous innovation and sustainability
- Roadmap infographic
- Call-to-action for partners

6.7 Documents and Resources Page

Purpose

Provide downloadable materials and visual resources.

Required Resource Categories

- Executive Portfolio
- Programme Prospectus

- Blueprint Summary
- Founder Roadmap
- Infographics
- Presentations
- Policy briefs, future
- Articles, future

Required Features

Each resource item shall include:

- title;
- short description;
- file type;
- download button;
- optional date;
- optional category.

For Phase 1, resources can be managed manually as static files.

6.8 Partnership Page

Purpose

Invite strategic partners to collaborate with GCRFIP.

Required Content Sections

- Why partner with GCRFIP?
- Partner categories:
  - government;
  - universities;
  - development partners;
  - NGOs;
  - private sector;
  - communities;
  - international organizations.
- Potential partner roles

- Benefits of partnership
- Expression-of-interest form
- Link to contact page

6.9 News / Insights Page

Purpose

Publish thought-leadership articles and updates.

Required Content Sections

- Article list
- Featured article
- Categories:
  - flood resilience;
  - climate intelligence;
  - community monitoring;
  - public health;
  - governance;
  - technology;
  - research.
- Individual article pages

For Phase 1, articles may be stored as markdown files.

6.10 Contact Page

Purpose

Allow visitors to contact the GCRFIP team.

Required Content Sections

- Contact form
- Email address
- LinkedIn/social media links

- Short message inviting collaboration
- Optional location or institutional affiliation

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7. Functional Requirements

FR-1: Responsive Website

The website shall be fully responsive across:

- desktop;
- tablet;
- mobile.

FR-2: Navigation

The website shall include a clear navigation menu with links to all major pages.

FR-3: Contact Form

The website shall include a contact form with the following fields:

- full name;
- organization;
- email address;
- subject;
- message;
- consent checkbox.

FR-4: Partnership Expression-of-Interest Form

The website shall include a partnership form with the following fields:

- full name;
- organization;
- role/title;
- email address;
- country;
- organization type;
- area of interest;
- proposed contribution;
- message;

- consent checkbox.

Organization type options:

- government;
- university/research institution;
- NGO/civil society;
- private sector;
- development partner;
- community organization;
- international organization;
- other.

Area of interest options:

- scientific research;
- community monitoring;
- technology development;
- public health;
- governance;
- capacity building;
- funding/investment;
- policy support;
- implementation partnership;
- other.

FR-5: Newsletter Signup

The website shall include a newsletter signup form with:

- name;
- email;
- optional organization;
- consent checkbox.

FR-6: Downloadable Resources

Users shall be able to download PDF files and visual resources.

FR-7: News / Insights

The website shall support publication of simple articles or updates.

FR-8: SEO Metadata

Each page shall include:

- title;
- meta description;
- Open Graph image;
- keywords where appropriate.

FR-9: Analytics

The website shall support analytics integration using:

- Google Analytics;
- Plausible;
- or another privacy-friendly analytics tool.

FR-10: Accessibility

The website shall follow basic accessibility best practices:

- readable contrast;
- alt text for images;
- keyboard navigability;
- semantic HTML;
- descriptive link labels;
- responsive typography.

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8. Non-Functional Requirements

Performance

The website shall load quickly and be optimized for performance.

Target:

- good Lighthouse performance score;
- optimized images;
- lazy loading for large visuals;
- minimal unnecessary scripts.

Security

The website shall:

- use HTTPS;
- avoid exposing API keys;
- validate form inputs;
- include spam protection for forms;
- use environment variables for sensitive configuration.

Maintainability

The code shall be clean, modular, and well-documented.

The developer shall organize content and components clearly.

Scalability

The website shall be designed so that future features can be added, including:

- database;
- CMS;
- partner portal;
- dashboard;
- event system;
- live map;
- document search.

Usability

The website shall be simple, professional, and easy to navigate.

Users should understand the purpose of GCRFIP within 10 seconds of landing on the Home page.

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9. Design Requirements

Visual Identity

The website shall use a professional Ghana-inspired colour palette.

Recommended colours:

- deep green;
- navy blue;
- gold/yellow;
- white;
- light grey;
- subtle red accents where appropriate.

Typography

Use modern, professional, readable fonts.

Recommended:

- headings: strong sans-serif;
- body: clean sans-serif;
- avoid decorative fonts for core text.

Design Style

The website should feel:

- credible;
- modern;
- institutional;
- inspiring;
- climate-focused;
- suitable for government and development partners.

Imagery

Use:

- Ghana cityscapes;
- flood resilience imagery;
- community participation;
- scientific research;
- technology and dashboards;
- maps;
- rivers and wetlands;
- professional infographics.

Avoid:

- overly dramatic disaster imagery;
- low-resolution images;
- generic stock visuals that do not fit Ghana or climate resilience.

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10. Content Requirements

The developer shall not create technical programme content independently.

Content shall be supplied by the project owner, including:

- programme text;
- documents;
- infographics;
- logos;
- images;
- article content;
- contact information.

The developer may assist with formatting, layout, and light editing for web readability.

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11. Forms and Data Handling

Since Phase 1 does not require a database, form submissions shall be handled through a third-party form service.

Required Form Behaviour

When a user submits a form:

- form validation shall occur;
- user shall see a success message;
- submission shall be sent to the designated email address;
- data shall be available in the form service dashboard;
- spam protection shall be applied.

Form Confirmation Message

Example:

"Thank you for contacting GCRFIP. Your message has been received, and we will respond as soon as possible."

Privacy

Forms shall include consent text such as:

"By submitting this form, you consent to GCRFIP using your information to respond to your inquiry."

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12. Deployment Requirements

The developer shall deploy the website to:

- Vercel, preferred for Next.js;
- or Netlify, acceptable alternative.

The deployment shall include:

- production build;
- custom domain setup support;
- HTTPS;
- environment variables;
- form service configuration;
- analytics setup;
- favicon;
- social sharing preview image.

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13. Suggested Folder Structure

For a Next.js implementation:

```
/app
/components
/content
/public
/public/documents
/public/images
/public/infographics
/styles
```

/lib
/data

Suggested component categories:

- Layout
- Header
- Footer
- Hero
- PillarCard
- ResourceCard
- PartnerCard
- CTASection
- FormComponents
- ArticleCard

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14. Future Phase Considerations

The developer should build Phase 1 in a way that allows future upgrades.

Possible Phase 2 features:

- database integration;
- CMS;
- partner portal;
- admin dashboard;
- live map interface;
- flood dashboard prototype;
- event registration;
- multilingual support;
- downloadable media kit;
- document analytics;
- newsletter automation.

Possible Phase 3 features:

- GIS flood risk viewer;
- real-time sensor data;
- interactive National Flood Intelligence Platform demo;

- institutional login;
- partner collaboration workspace;
- AI-supported document search;
- open data portal.

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15. Acceptance Criteria

The website shall be considered complete when:

- all agreed pages are implemented;
- design is responsive across desktop, tablet, and mobile;
- navigation works correctly;
- all forms submit successfully;
- downloadable documents are accessible;
- SEO metadata is configured;
- analytics is installed;
- images are optimized;
- site is deployed publicly;
- custom domain is connected, if available;
- stakeholder review feedback is addressed.

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16. Initial Phase Deliverables

The developer shall deliver:

- complete website source code;
- deployed website URL;
- deployment documentation;
- form service configuration;
- instructions for updating content;
- instructions for adding documents;
- basic maintenance guide;
- list of third-party services used.

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17. Project Success Definition

The Phase 1 GCRFIP website will be successful if it provides a professional, credible, and persuasive online presence that allows stakeholders to understand the programme, access key documents, and express interest in collaboration.

The website should make GCRFIP appear ready for serious engagement with government institutions, universities, development partners, NGOs, private sector actors, and international collaborators.

Final Statement

The Phase 1 GCRFIP website is not merely a communication tool.

It is the digital front door of the programme.

It must communicate credibility, vision, professionalism, and readiness for partnership.

The website should leave visitors with one clear impression:

GCRFIP is a serious, well-designed, nationally important climate resilience programme worthy of collaboration, support, and investment.

4. Website Information Architecture

To ensure long-term scalability, usability, and maintainability, the GCRFIP website shall adopt a hierarchical information architecture consisting of **Main Pages (Level 1)**, **Section Pages (Level 2)**, and **Content Pages (Level 3)**.

This structure minimises navigation complexity while allowing the website to grow as new programmes, publications, tools, datasets, and partnerships are added.

The website shall initially present **eight primary navigation items** on the main navigation bar.

4.1 Main Navigation Structure

The website shall contain the following primary navigation menu:

- Home
- About
- Programme
- Research
- Resources
- News
- Get Involved
- Contact

Each primary menu shall serve as a gateway to related sub-pages.

4.2 Site Hierarchy

```
HOME
│
├── About
│   ├── About GCRFIP
│   ├── Vision & Mission
│   ├── Why Ghana Needs GCRFIP
│   ├── Leadership
│   ├── Governance
│   ├── Advisory Board
│   └── Strategic Partners
│
```

```
│
├── Programme
│   ├── Programme Overview
│   ├── Executive Portfolio
│   ├── Programme Prospectus
│   ├── Blueprint
│   ├── National Roadmap
│   ├── Implementation Strategy
│   └── Six Strategic Pillars
│        ├── Pillar 1
│        ├── Pillar 2
│        ├── Pillar 3
│        ├── Pillar 4
│        ├── Pillar 5
│        └── Pillar 6
│
├── Research
│   ├── Research Themes
│   ├── Publications
│   ├── Innovation Lab
│   ├── Flood Intelligence Platform
│   ├── Digital Twin
│   ├── Open Data
│   ├── Student Opportunities
│   └── Research Projects
│
├── Resources
│   ├── Reports
│   ├── Policy Briefs
│   ├── Infographics
│   ├── Maps
│   ├── Videos
│   ├── FAQs
│   ├── Downloads
│   └── Training Materials
│
├── News
│   ├── Latest News
│   ├── Events
│   ├── Workshops
│   ├── Conferences
│   ├── Press Releases
│   └── Blog
│
├── Get Involved
│   ├── Become a Partner
│   ├── Join the Founding Consortium
│   ├── Research Collaboration
│   ├── Volunteer
│   ├── Sponsorship & Funding
```

```
│      ├── Careers (Future)
│      └── Newsletter Subscription
│
└── Contact
       ├── Contact Form
       ├── Office Information
       ├── Interactive Map
       ├── Social Media
       └── General Enquiries
```

4.3 Homepage Functional Requirements

The homepage shall function as the digital gateway to the entire programme and shall communicate the value proposition of GCRFIP within the first 30 seconds of a visitor's arrival.

The homepage shall include:

Hero Section

- High-quality background image or video.
- Programme title.
- Vision statement.
- Primary call-to-action buttons:
    - Learn More
    - Explore the Blueprint
    - Become a Partner

National Flood Challenge Overview

- Brief explanation of Ghana's flood challenges.
- Supporting infographic.
- Key statistics.

About GCRFIP

- Short programme overview.
- Link to the About section.

Six Strategic Pillars

- Interactive cards introducing each pillar.
- Direct links to pillar pages.

Programme Highlights

- Executive Portfolio.
- Prospectus.
- Blueprint.
- National Roadmap.

Featured Research

- Recent publications.
- Innovation projects.
- Research highlights.

Latest News & Events

- Dynamic news feed.
- Upcoming events.
- Workshops.

Strategic Partners

- Partner logos.
- Link to partnership page.

Call to Action

- Join the Consortium.
- Become a Research Partner.
- Subscribe to updates.

Footer

26. Contact information.
27. Quick links.
28. Social media.
29. Copyright.
30. Privacy Policy.
31. Terms of Use.

4.4 Navigation Requirements

The website shall support:

- Responsive top navigation.
- Sticky navigation bar.
- Multi-level dropdown menus.
- Breadcrumb navigation on internal pages.
- Global search functionality.
- Mobile-friendly navigation menu.
- Footer navigation.
- Quick-access links to key resources.

4.5 Future Expansion

The website architecture shall be designed to accommodate future expansion without requiring structural redesign.

Future modules may include:

Secure Login Portal

Role-based access for:

- Researchers
- Government agencies
- Programme administrators
- Consortium members
- Community volunteers

National Flood Intelligence Platform

Future integration of:

- Operational dashboards
- Flood forecasts
- Live sensor data
- GIS maps
- Digital Twin visualisations
- Decision-support tools

Research Portal

Future capabilities:

- Dataset repository
- Publication management
- Project collaboration
- Research proposal submission
- Grant management

Community Portal

Future capabilities:

- Citizen reporting
- Flood observation submissions
- Volunteer management
- Community training
- Local preparedness resources

4.6 Information Architecture Principles

The website shall adhere to the following design principles:

- Simple and intuitive navigation.
- Logical grouping of related content.
- Minimal top-level navigation items.
- Scalable architecture supporting future growth.
- Clear separation between public information and secure operational services.
- Responsive design for desktop, tablet, and mobile devices.
- Compliance with recognised accessibility standards.
- Optimised performance and search engine visibility.
- Consistent branding and visual identity across all pages.

4.7 Design Philosophy

The GCRFIP website shall be developed as more than a public-facing information portal. It shall serve as the programme’s digital headquarters, providing a scalable foundation that evolves from an awareness and partnership platform in Phase 1 into a comprehensive national resilience ecosystem in later phases.

The architecture shall support the gradual integration of scientific research, operational intelligence, collaboration tools, public engagement services, and secure digital applications without requiring fundamental redesign. This phased approach ensures long-term sustainability while enabling incremental development aligned with programme growth and available resources.